

P2P Dynamic Port Switching

A Moving Target Defense Approach

Students

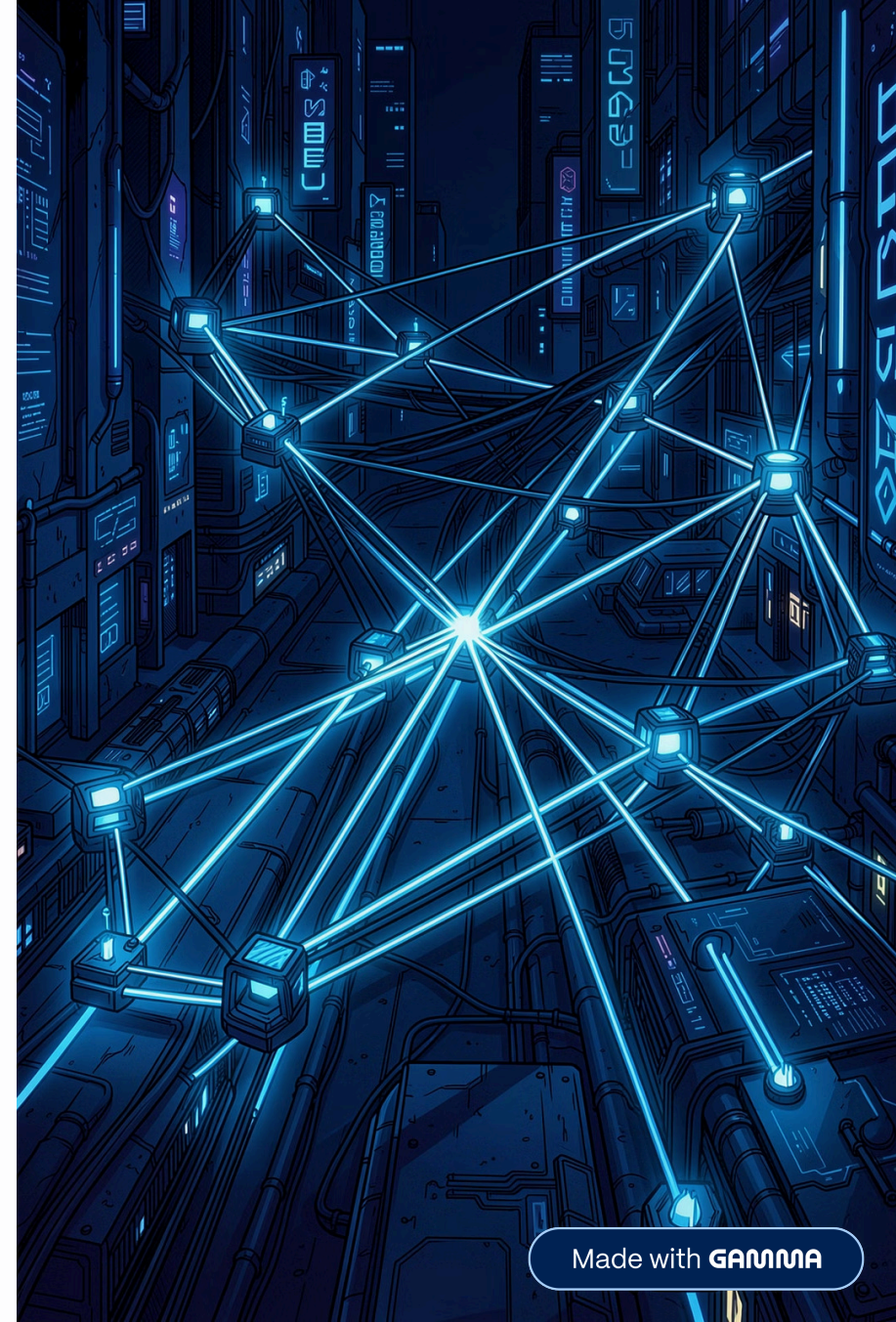
Eren Abdullah Beşik - 202128019

Şahin Erşan - 202128002

Berfin Duru Alkan - 202228005

Advisor

Atila Bostan from Computer Engineering Department



Problem: Static Ports Are Easy Targets

Predictability

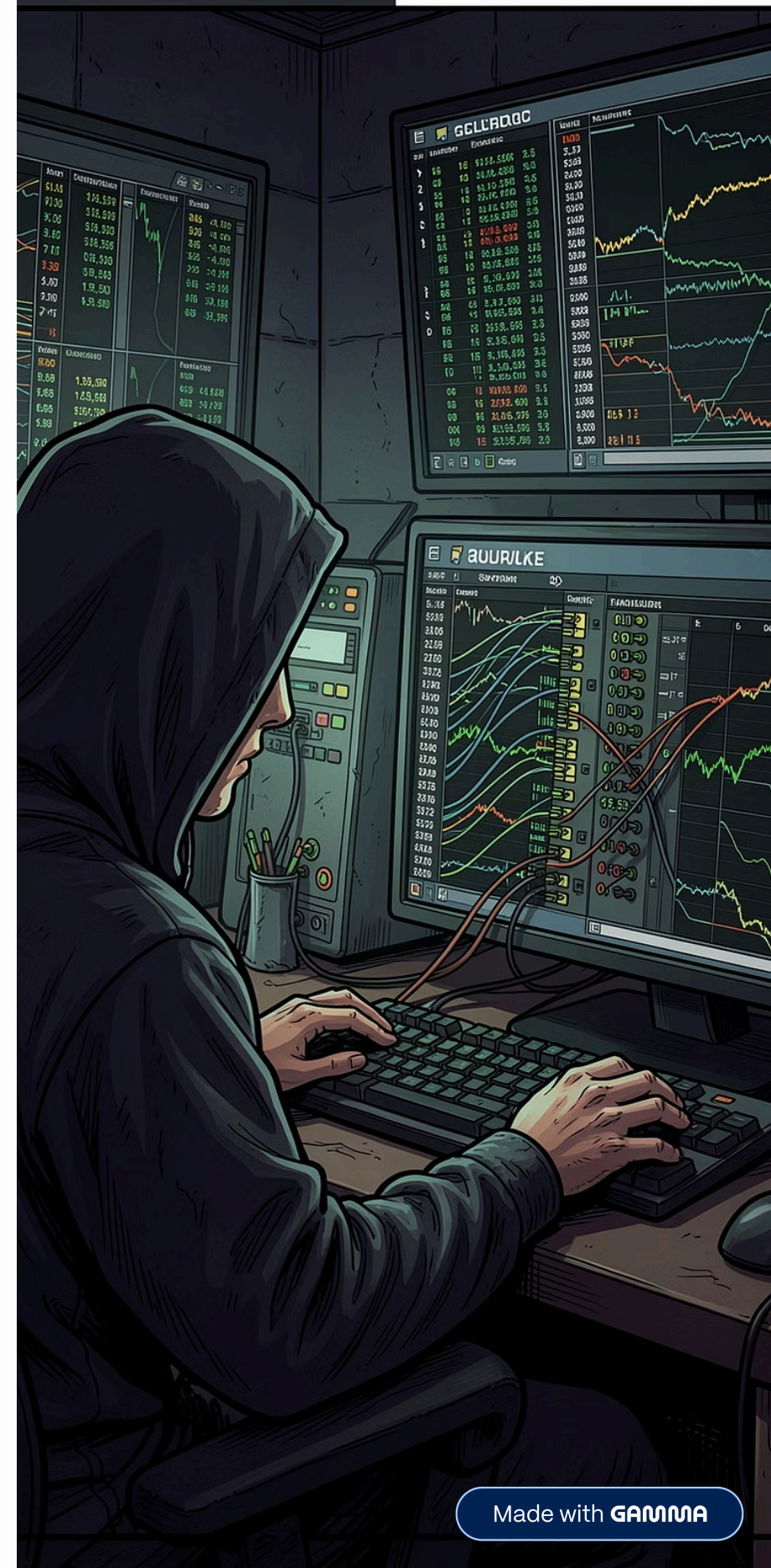
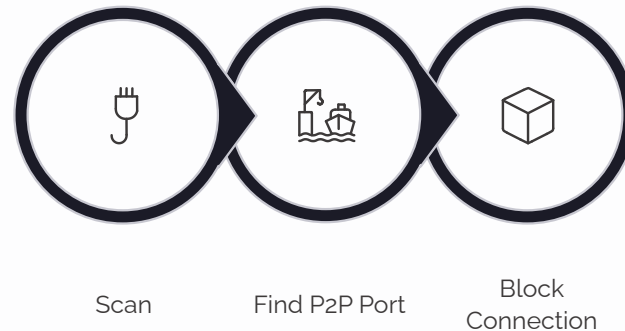
Traditional P2P systems use fixed port numbers — attackers can easily identify these ports.

Single Point of Vulnerability

Once a port is identified, all communication can be monitored, filtered, or completely blocked.

Firewall Exposure

Restrictive firewalls and attackers can immediately block known static ports.



Inspiration: Military Frequency Hopping

Adapting secure communication ideas from military origins to P2P networks.



Military Frequency Hopping

Military radio systems use frequency hopping technology to secure communications. This technique makes it harder for enemies to detect, jam, or intercept signals.

The radio signal **rapidly switches among many different frequencies on a periodic basis**, avoiding static targets.



Adaptation: Dynamic P2P Port Switching

We applied this proven security concept to P2P networks. Instead of static frequencies, our system **dynamically changes port numbers for P2P connections**.

This approach makes it extremely difficult for adversaries to detect or block a P2P port, creating a defense layer similar to the security level provided by frequency hopping in military operations.

Our Solution: Moving Target Defense (MTD)

"If the target is always moving, it becomes almost impossible to hit."



Dynamic Port Hopping

Every N seconds, the system switches to a new port — leaving no chance for the attacker to learn it.



SHA-256 Deterministic Computation

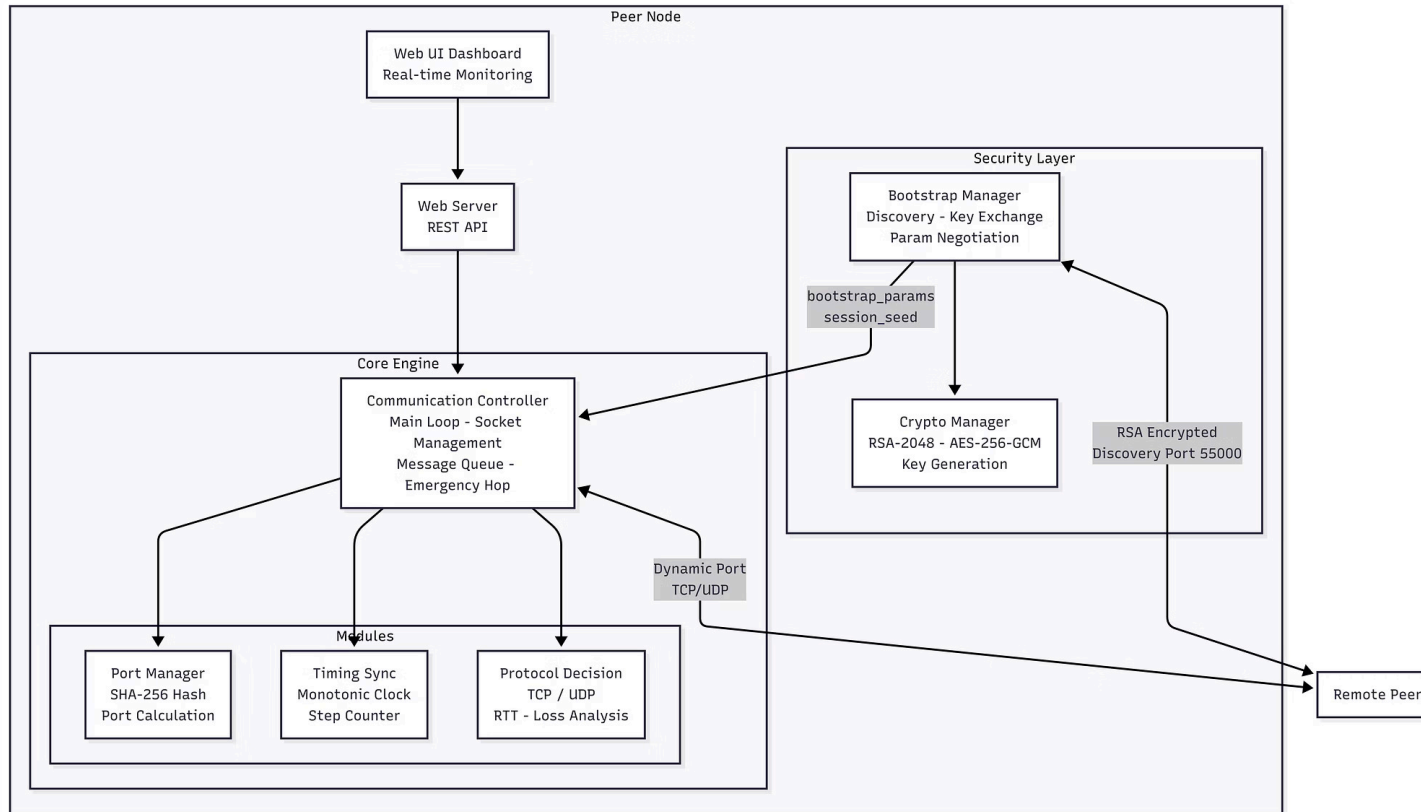
Ports are computed independently using a shared secret parameter; no extra messaging is required.



Automatic TCP/UDP Switching

The protocol is automatically selected at each step based on network conditions.

System Architecture



Bootstrap Manager

Initial handshake with RSA + AES

Port Manager

SHA-256-based port calculation

Timing Sync

Monotonic clock synchronization

Protocol Decision

TCP / UDP / AUTO selection

Stage 1: Secure Bootstrapping (Key Exchange)



This one-time handshake is complete, after which the discovery port is completely disabled. Now both sides share the same session parameters and can independently start the port-hopping process.



Stage 2: SHA-256-Based Port Hopping

Port Calculation Formula

```
hash = SHA-256(bootstrap_params + step  
+ peer_id)  
port = min_port + (hash mod  
(max_port - min_port + 1))
```

Each peer calculates **2 ports** for each step: its own local port and the other side's port.

i Example — Step 5: Peer A
→ local=25000,
peer=27500 | Peer B →
local=27500, peer=25000

Deterministic

The same parameters always produce the same port.

Unpredictable

The port sequence is cryptographically secure without a secret seed.

Collision-Safe

The hash space minimizes port collisions.

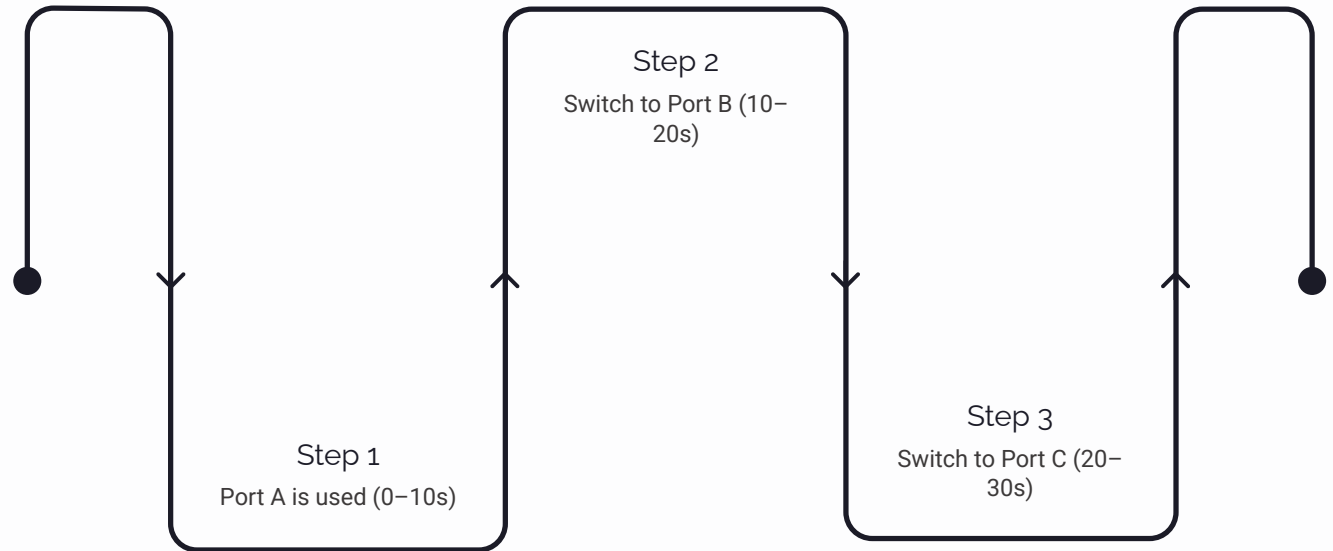
Monotonic Time Step Synchronization

Step Calculation

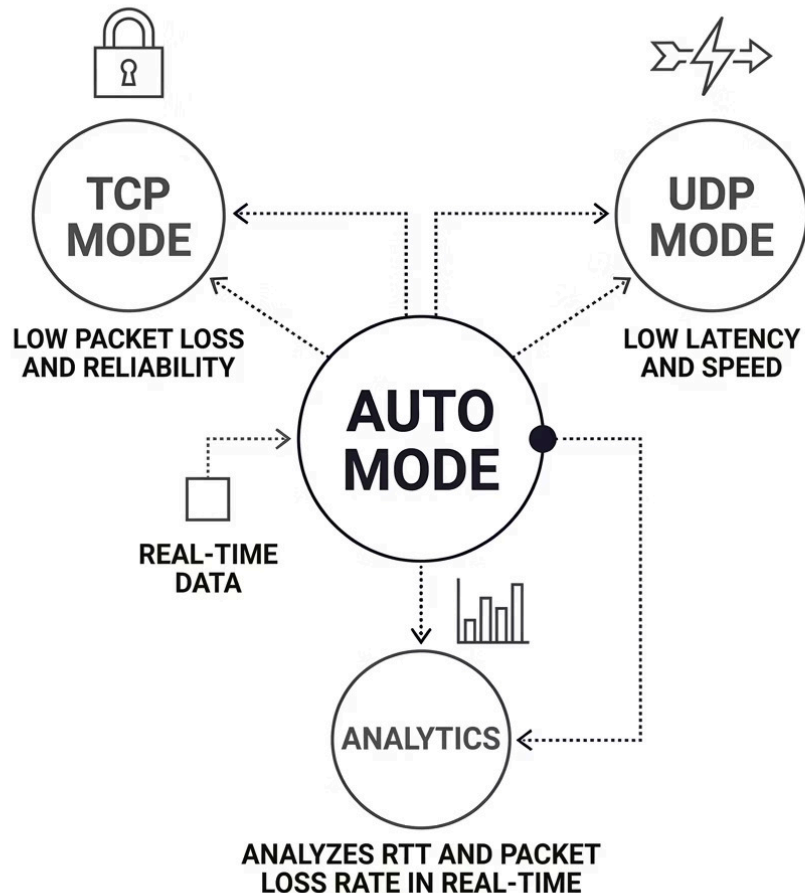
```
step = floor((now - epoch) / interval)
```

The session **epoch** agreed during bootstrapping is used as the reference. The default interval is **10 seconds**.

- ✓ Using a monotonic clock provides perfect synchronization independent of NTP updates or manual clock changes.



Adaptive Protocol Switching: TCP ↔ UDP



AUTO Mode Decision Logic

→ RTT Measurement

Real-time round-trip time is monitored using ACK timestamps.

→ Packet Loss Rate

The percentage of failed transmissions is calculated at each step.

→ Protocol Selection

High loss → TCP (reliability) | Low latency → UDP (speed)

Emergency Jump: Active Attack Recovery

Scenario

The attacker blocks the active port in the middle of the current port interval. Instead of waiting for the next step, the system automatically initiates recovery within **3 seconds**.

Emergency Port Calculation

$$\text{emergency_port} = \text{SHA-256}(\text{params} + \text{step} + \text{"emergency"})$$

⚠ If no successful DATA or ACK is received for 3 seconds, the emergency jump is triggered.

